

University of Kelaniya

Faculty of Computing and Technology



Internet of Things - CTEC 32023

Mini-Project

Off-Grid Smart Agriculture Node

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01. Project Objective

This project is about creating an agriculture monitoring system that works with internet connected devices. The system will use tools to check the soil and air. It will look at how wet the soil's how hot or cold it is, how humid it is and how much light the plants are getting. This information will be sent to a website where farmers can see how their crops are doing from far away using their phone or computer.

The project also wants to help farmers use energy by using sunlight to power the system. This way the smart agriculture monitoring system and the farmers can use energy as the main power source, for the smart agriculture monitoring system.

02. Problem Description

Agriculture is an important industry in Sri Lanka and many other countries. Farmers need to know about the soil and weather conditions to grow crops and produce more.

However, many farms are in areas where there is no easy access to electricity and internet. Farmers usually have to check the soil moisture and crop conditions by themselves. This is time-consuming and can cause problems like:

- Overwatering or under watering crops
- Wasting water
- Spending more on labor
- Getting crop yields
- Finding it hard to regularly check on the fields

To fix these issues we need a smart system to monitor farms. The Off-Grid Smart Agriculture Node that is being proposed will automatically gather data on the environment. Send it to the cloud. Farmers can get this information from anywhere. Make smarter choices about watering and managing crops.

Farmers will use this information to improve their crop management and make decisions, about irrigation.

The Off-Grid Smart Agriculture Node will help farmers to increase crop productivity and reduce water wastage.

04. Hardware Components

Component	Purpose
ESP32 node mcu	Main controller
SIM900A SIM Module	Internet connectivity
Soil Moisture Sensor	Soil condition monitoring
DHT22 Sensor	Temperature and humidity measurement
LDR Sensor	Light intensity monitoring
Solar Panel (5V–6V)	Renewable energy source
18650 Rechargeable Battery	Energy storage
TP4056 Charging Module	Battery charging management
Buck/Boost Converter	Voltage regulation
470µF–1000µF Capacitor	GSM power stabilization
Breadboard & Jumper Wires	Circuit connections

Software and Technologies

- Arduino IDE
- ESP32 Programming
- TinyGSM Library
- ThingSpeak / Blynk Cloud Platform
- MQTT / HTTP Protocol
- EasyEDA
- Android Studio (Optional Mobile Application)

04. System Working Process

- The solar panel generates electricity during the daytime.
- The battery stores energy for continuous operation.
- Sensors collect environmental data from the farm.
- The ESP32 reads and processes sensor values.
- The SIM800L module connects to the internet using a mobile network.
- Sensor data is uploaded to the cloud platform.
- Farmers can view real-time data using a mobile phone or computer.
- The ESP32 enters deep sleep mode to save battery power and wakes up periodically to send new readings.

05. Expected Outcomes

After completing the project, the system will be able to:

- Monitor soil moisture levels automatically.
- Measure temperature and humidity in real time.
- Monitor sunlight intensity.
- Send sensor data to a cloud platform.
- Allow remote monitoring through a smartphone.
- Operate using solar power without grid electricity.
- Reduce water wastage and improve irrigation management.
- Support smart farming practices in rural areas.

06. Innovation of the Project

This project is really cool because it can work on its own without being connected to any power lines. It uses energy from the sun to run, which is a difference from other systems that need electricity and internet to work.

The system uses power and sends information through GSM, which is like a phone signal. This makes it perfect for farms that're far away from cities where you cannot get electricity or internet. The project is also very affordable. It does not waste energy, which is good, for the environment. This solar energy project is a way to help farmers in remote areas.

8. Conclusion

The Off-Grid Smart Agriculture Node is a useful tool for farming. It is a way for farmers to keep an eye on things from far away. The Off-Grid Smart Agriculture Node uses energy from the sun, which's a good thing for the environment. It helps farmers take care of water better it reduces the amount of work they have to do and it helps them farm in a way that is good, for the earth. This project shows how IOT technology can be used to solve problems that farmers face every day and it does this in a way that is affordable and works well.